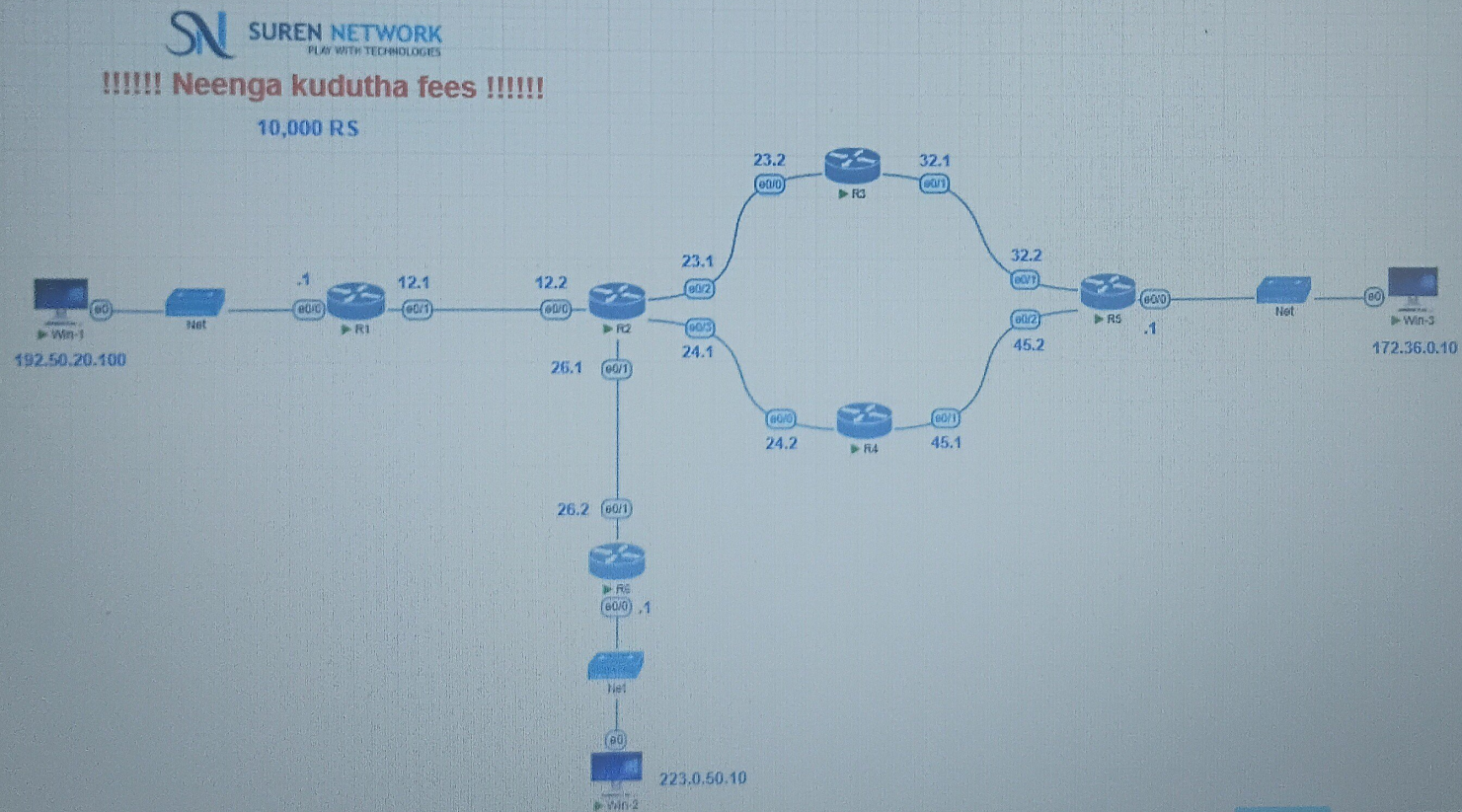




R1

```
Router#  
Router#  
Router#sh ip int brief  
Interface IP-Address OK? Method Status Prot  
Loopback0 192.50.20.1 YES manual up  
Ethernet0/0 12.0.0.1 YES manual up  
Ethernet0/1 unassigned YES unset administratively down down  
Ethernet0/2 unassigned YES unset administratively down down  
Ethernet0/3 unassigned YES unset administratively down down
```

```
Router#sh ip route  
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP  
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area  
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2  
E1 - OSPF external type 1, E2 - OSPF external type 2  
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2  
ia - IS-IS inter area, * - candidate default, U - per-user static route  
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP  
a - application route  
+ - replicated route, % - next hop override, p - overrides from PfR
```

Gateway of last resort is not set

```
12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks  
C 12.0.0.0/8 is directly connected, Ethernet0/1  
L 12.0.0.1/32 is directly connected, Ethernet0/1  
S 23.0.0.0/8 [1/0] via 12.0.0.2  
S 24.0.0.0/8 [1/0] via 12.0.0.2  
S 26.0.0.0/8 [1/0] via 12.0.0.2  
S 32.0.0.0/8 [1/0] via 12.0.0.2  
S 45.0.0.0/8 [1/0] via 12.0.0.2  
S 172.36.0.0/16 [1/0] via 12.0.0.2  
192.50.20.0/24 is variably subnetted, 2 subnets, 2 masks  
C 192.50.20.0/24 is directly connected, Ethernet0/0  
L 192.50.20.1/32 is directly connected, Ethernet0/0  
S 223.0.50.0/24 [1/0] via 12.0.0.2
```

```
Router#  
Router#  
Router#  
Router#
```

30°C

R2

Router>enable

Router#sh ip int brief

Interface	IP-Address	OK?	Method	Status	Prot
o001					
Ethernet0/0	12.0.0.2	YES	manual	up	up
Ethernet0/1	26.0.0.1	YES	manual	up	up
Ethernet0/2	23.0.0.1	YES	manual	up	up
Ethernet0/3	24.0.0.1	YES	manual	up	up

Router#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

C 12.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 12.0.0.0/8 is directly connected, Ethernet0/0
L 12.0.0.2/32 is directly connected, Ethernet0/0
C 23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 23.0.0.0/8 is directly connected, Ethernet0/2
L 23.0.0.1/32 is directly connected, Ethernet0/2
C 24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
L 24.0.0.0/8 is directly connected, Ethernet0/3
L 24.0.0.1/32 is directly connected, Ethernet0/3
C 26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks

--More--

R3

```
Router>
Router>enable
Router#sh ip int brief
Interface      IP-Address      OK? Method Status      Prot
-----
ooc1
Ethernet0/0      23.0.0.2        YES manual up          up
Ethernet0/1      32.0.0.1        YES manual up          up
Ethernet0/2      unassigned      YES unset  administratively down down
Ethernet0/3      unassigned      YES unset  administratively down down
```

Router#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    12.0.0.0/8 [1/0] via 23.0.0.1
    23.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    23.0.0.0/8 is directly connected, Ethernet0/0
L    23.0.0.2/32 is directly connected, Ethernet0/0
S    24.0.0.0/8 [1/0] via 23.0.0.1
S    26.0.0.0/8 [1/0] via 23.0.0.1
    32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.1/32 is directly connected, Ethernet0/1
S    45.0.0.0/8 [1/0] via 32.0.0.2
--More--
```


R4

```
Router>
Router>enable
Router#sh ip int b
Interface                IP-Address      OK? Method Status      Prot
-----
Gig0/0                    24.0.0.2        YES manual up          up
Gig0/1                    45.0.0.1        YES manual up          up
Gig0/2                    unassigned      YES unset  administratively down down
Gig0/3                    unassigned      YES unset  administratively down down
```

```
Router#sh ip route
```

```
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2
       i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
       ia - IS-IS inter area, * - candidate default, U - per-user static route
       o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
       a - application route
       + - replicated route, % - next hop override, p - overrides from PfR
```

```
Gateway of last resort is not set
```

```
S    12.0.0.0/8 [1/0] via 24.0.0.1
S    23.0.0.0/8 [1/0] via 24.0.0.1
     24.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     24.0.0.0/8 is directly connected, Ethernet0/0
L     24.0.0.2/32 is directly connected, Ethernet0/0
S    26.0.0.0/8 [1/0] via 24.0.0.1
     45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C     45.0.0.0/8 is directly connected, Ethernet0/1
L     45.0.0.1/32 is directly connected, Ethernet0/1
S    172.36.0.0/16 [1/0] via 45.0.0.2
```

```
--More--
```


R5

```
ccol
Ethernet0/0      172.36.0.1      YES manual up      up
Ethernet0/1      32.0.0.2      YES manual up      up
Ethernet0/2      45.0.0.2      YES manual up      up
Ethernet0/3      unassigned    YES unset   administratively down down
```

Router#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
E1 - OSPF external type 1, E2 - OSPF external type 2
i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
ia - IS-IS inter area, * - candidate default, U - per-user static route
o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
a - application route
+ - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```
S    12.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    23.0.0.0/8 [1/0] via 32.0.0.1
S    24.0.0.0/8 [1/0] via 45.0.0.1
S    26.0.0.0/8 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
32.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    32.0.0.0/8 is directly connected, Ethernet0/1
L    32.0.0.2/32 is directly connected, Ethernet0/1
45.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    45.0.0.0/8 is directly connected, Ethernet0/2
L    45.0.0.2/32 is directly connected, Ethernet0/2
172.36.0.0/16 is variably subnetted, 2 subnets, 2 masks
C    172.36.0.0/16 is directly connected, Ethernet0/0
L    172.36.0.1/32 is directly connected, Ethernet0/0
S    192.50.20.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
S    223.0.50.0/24 [1/0] via 45.0.0.1
      [1/0] via 32.0.0.1
```

Router#

Router#

30°C

Router>

Router>enable

Router#sh ip int b

Interface	IP-Address	OK?	Method	Status	Prot
Ethernet0/0	223.0.50.1	YES	manual	up	up
Ethernet0/1	26.0.0.2	YES	manual	up	up
Ethernet0/2	unassigned	YES	unset	administratively down	down
Ethernet0/3	unassigned	YES	unset	administratively down	down

Router#sh ip route

Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
 D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
 N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
 E1 - OSPF external type 1, E2 - OSPF external type 2
 i - IS-IS, su - IS-IS summary, L1 - IS-IS level-1, L2 - IS-IS level-2
 ia - IS-IS inter area, * - candidate default, U - per-user static route
 o - ODR, P - periodic downloaded static route, H - NHRP, l - LISP
 a - application route
 + - replicated route, % - next hop override, p - overrides from PfR

Gateway of last resort is not set

```

S    12.0.0.0/8 [1/0] via 26.0.0.1
S    23.0.0.0/8 [1/0] via 26.0.0.1
S    24.0.0.0/8 [1/0] via 26.0.0.1
    26.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C    26.0.0.0/8 is directly connected, Ethernet0/1
L    26.0.0.2/32 is directly connected, Ethernet0/1
S    32.0.0.0/8 [1/0] via 26.0.0.1
S    45.0.0.0/8 [1/0] via 26.0.0.1
S    172.36.0.0/16 [1/0] via 26.0.0.1
S    192.50.20.0/24 [1/0] via 26.0.0.1
    223.0.50.0/24 is variably subnetted, 2 subnets, 2 masks
C    223.0.50.0/24 is directly connected, Ethernet0/0
L    223.0.50.1/32 is directly connected, Ethernet0/0

```

Router#

Router#

✓ New tab

← ↻ https://lab

QEMU (Win-1)



C:\WINDOWS\system32\cmd.exe

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=4ms TTL=125

Reply from 223.0.50.10: bytes=32 time=2ms TTL=125

Reply from 223.0.50.10: bytes=32 time=1ms TTL=125

Reply from 223.0.50.10: bytes=32 time=1ms TTL=125

Ping statistics for 223.0.50.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 4ms, Average = 2ms

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Reply from 172.36.0.10: bytes=32 time=1ms TTL=124

Reply from 172.36.0.10: bytes=32 time=1ms TTL=124

Reply from 172.36.0.10: bytes=32 time=2ms TTL=124

Ping statistics for 172.36.0.10:

Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),

Approximate round trip times in milli-seconds:

Minimum = 1ms, Maximum = 2ms, Average = 1ms

192.5

C:\Documents and Settings\user>_

QEMU (Win-2)



C:\WINDOWS\system32\cmd.exe

```
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
Reply from 192.50.20.100: bytes=32 time=1ms TTL=125
Reply from 192.50.20.100: bytes=32 time=2ms TTL=125
```

Ping statistics for 192.50.20.100:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 1ms, Maximum = 2ms, Average = 1ms
```

C:\Documents and Settings\user>

C:\Documents and Settings\user>ping 172.36.0.10

Pinging 172.36.0.10 with 32 bytes of data:

```
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=2ms TTL=124
Reply from 172.36.0.10: bytes=32 time=3ms TTL=124
```

Ping statistics for 172.36.0.10:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 3ms, Average = 2ms
```

C:\Documents and Settings\user>_

192.5

QEMU (Win-3)

C:\WINDOWS\system32\cmd.exe

```
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124
Reply from 192.50.20.100: bytes=32 time=1ms TTL=124
Reply from 192.50.20.100: bytes=32 time=2ms TTL=124

Ping statistics for 192.50.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 2ms, Average = 1ms

C:\Documents and Settings\user>
C:\Documents and Settings\user>ping 223.0.50.10

Pinging 223.0.50.10 with 32 bytes of data:

Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=1ms TTL=124
Reply from 223.0.50.10: bytes=32 time=3ms TTL=124
Reply from 223.0.50.10: bytes=32 time=2ms TTL=124

Ping statistics for 223.0.50.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 3ms, Average = 1ms
192.5 C:\Documents and Settings\user>_
```